

Temperature P2 questions 5054

Compiled by: Mustafa Asif

10. Temperature

Content

10.1 Principles of thermometry

10.2 Practical thermometers

Learning outcomes

Candidates should be able to:

(a) explain how a physical property which varies with temperature may be used for the measurement of

temperature and state examples of such properties.

(b) explain the need for fixed points and state what is meant by the *ice point* and *steam point*.

(c) discuss sensitivity, range and linearity of thermometers.

(d) describe the structure and action of liquid-in-glass thermometers (including clinical) and of a thermocouple thermometer, showing an appreciation of its use for measuring high temperatures and

those which vary rapidly.

(e) describe and explain how the structure of a liquid-in-glass thermometer affects its sensitivity, range and

linearity.

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O/N18/21/Q5

A nurse uses a mercury-in-glass, clinical thermometer to measure the temperature of a patient.

Fig. 5.1 shows the structure of a clinical thermometer.

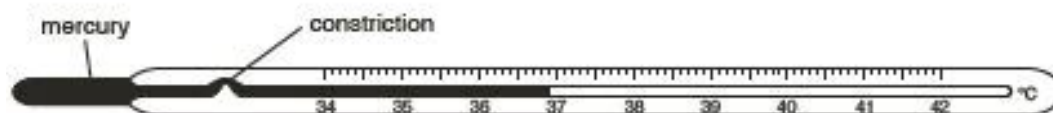


Fig. 5.1

(a) The clinical thermometer has a small range and a high sensitivity.

(i) State what is meant by

1. *range*,

.....
..... [1]

2. *sensitivity*.

.....
..... [1]

(ii) State and explain one feature of this clinical thermometer that produces a high sensitivity.

.....
.....
.....
..... [2]

(b) State the purpose of the constriction in the clinical thermometer.

.....
..... [1]

(c) Explain, in terms of the mercury molecules, how an increase in temperature produces an increased reading on the thermometer.

.....
.....
.....
.....
..... [3]

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O/N17/21/Q4

Fig. 4.1 shows a mercury-in-glass thermometer made using a glass capillary tube.

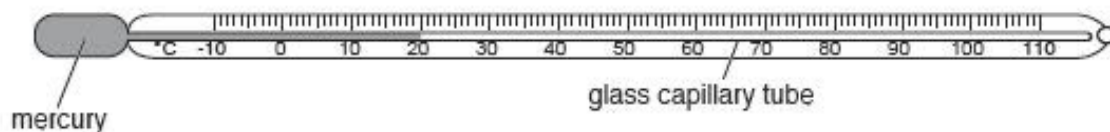


Fig. 4.1

- (a) Describe how to check that the 100°C mark is in the correct position on the thermometer.

.....

.....

.....

..... [3]

- (b) Some thermometers contain ethanol instead of mercury.

A second glass capillary tube is identical to the tube in Fig. 4.1, except that it has no markings on the glass. This tube is used to make an ethanol-in-glass thermometer. The volume of the ethanol in this thermometer is equal to the volume of mercury in the thermometer in Fig. 4.1.

The thermal expansion of ethanol is more than that of mercury.

State and explain how the maximum possible range of the ethanol-in-glass thermometer differs from the range of the thermometer in Fig. 4.1.

.....

.....

..... [2]

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M/J17/21/Q11(c)

(c) A thermocouple thermometer is another type of thermometer.

- (i) Describe the structure of a thermocouple thermometer. You may draw a diagram if you wish.

.....

.....

.....

.....

.....[3]

- (ii) State two advantages of a thermocouple thermometer over a liquid-in-glass thermometer.

1.

.....

2.

.....

.....[2]

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0/N16/21/Q4

- (a) Equal volumes of air, copper and water are heated from 20°C to 80°C . They expand by different amounts.

Draw a line from each box on the left-hand side to show which expands the most and which expands the least.

expands the most	air
	copper
expands the least	water

[1]

- (b) Fig. 4.1 shows a mercury thermometer.



Fig. 4.1

This thermometer has a suitable range for use in laboratory experiments, but has a low sensitivity for some experiments.

- (i) With reference to this thermometer, describe what is meant by

1. *range*,

.....
.....
.....[1]

2. *low sensitivity*.

.....
.....
.....[1]

- (ii) State one change in the design of a thermometer that increases its sensitivity.

.....
.....[1]

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0/N15/22/Q5

All thermometers use the value of a physical property to measure temperature.

- (a) (i) State what makes a physical property suitable for the measurement of temperature.

.....
.....
..... [1]

- (ii) State two properties that are used for the measurement of temperature.

1.
2. [2]

- (b) When a thermometer is calibrated, two fixed points are used.

- (i) One fixed point is the ice point. State what is meant by the *ice point*.

.....
.....
..... [1]

- (ii) Explain how the fixed points are used when calibrating a thermometer.

.....
.....
.....
.....
.....
.....
.....
..... [2]

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0/N13/21/Q5

- (a) Draw a labelled diagram to show the structure of a liquid-in-glass clinical thermometer.

[2]

- (b) A clinical thermometer is used to measure the temperature of a patient. When removed, the thermometer cools but it continues to show the maximum temperature reached.

Explain how this happens.

.....

.....

..... [2]

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M/J13/21/Q4

- 4 (a) Fig. 4.1 shows a laboratory thermometer.

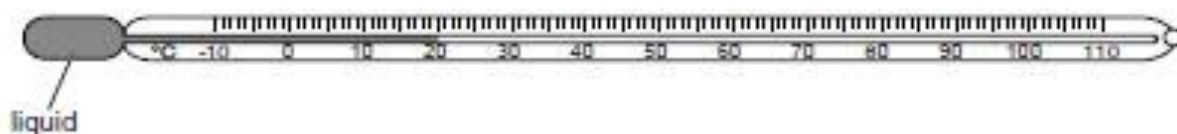


Fig. 4.1

- (I) State the range of the thermometer shown in Fig. 4.1.

..... [1]

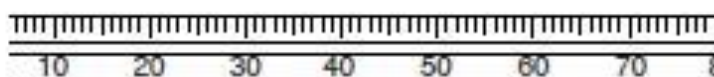
- (II) Describe how Fig. 4.1 shows that the thermometer has a linear scale.

.....
.....
..... [1]

- (III) The manufacturer of the thermometer changes the design to make a more sensitive thermometer.

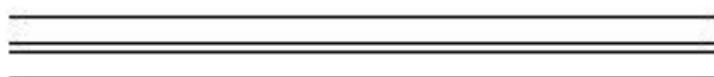
Fig. 4.2 shows part of the scale on the original thermometer.

On Fig. 4.3, draw a more sensitive scale.



part of the scale
on the original
thermometer

Fig. 4.2



your scale showing
a more sensitive
thermometer

Fig. 4.3

[1]

- (b) State one other property, apart from volume or expansion, which can be used to measure temperature.

..... [1]

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0/N12/21/Q2

- 2 A thermocouple thermometer is made from two different metals and a voltmeter. One junction of the thermocouple thermometer is placed in a beaker containing a mixture of water and ice. The other junction is placed inside a high-temperature furnace, as shown in Fig. 2.1.

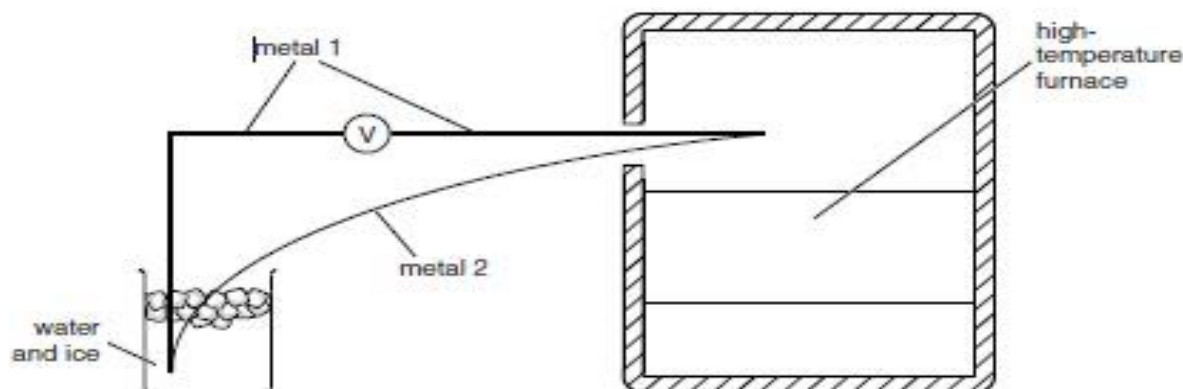


Fig. 2.1

- (a) The thermocouple thermometer has a linear output over the operating temperature range of the furnace.

(i) State what, for a thermocouple thermometer, is meant by a *linear output*.

..... [1]

- (ii) The junction inside the furnace is at temperature T . Fig. 2.2 shows the reading V of the voltmeter at two values of T that are within the operating temperature range of the furnace.

$T/^{\circ}\text{C}$	V/mV
750	6.20
800	7.70

Fig. 2.2

When the furnace is operating at a high temperature, the furnace door is opened and the reading of the voltmeter falls to 6.80 mV.

Calculate the temperature of the junction inside the furnace.

temperature = [2]

- (b) State why a liquid-in-glass thermometer is not used to measure the temperature inside the furnace.

..... [1]

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0/N11/21/Q3

Fig. 3.1 shows the flame of a Bunsen burner.

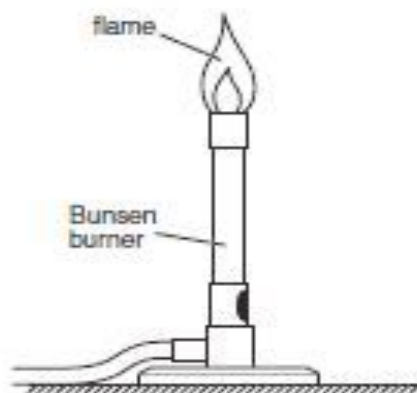


Fig. 3.1

- (a) A thermocouple thermometer contains a voltmeter. It is used to measure the temperature in the flame of the Bunsen burner.

(i) On Fig. 3.1, draw a labelled diagram of the structure of the thermocouple thermometer as it measures the temperature in the flame. [2]

(ii) Describe how the fixed points and the reading of the voltmeter are used to determine the value of the temperature in the flame.

.....

.....

.....

.....[2]

- (b) The thermocouple thermometer is used to measure the temperature in the flame because it can measure very high temperatures. State and explain one other advantage of using a thermocouple thermometer to determine temperature.

.....

.....

.....[2]

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M/J10/21/Q9

9 Fig. 9.1 shows a laboratory thermometer.

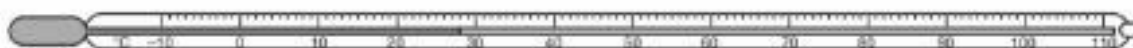


Fig. 9.1

- (a) (i) State the range of the thermometer.
- [1]
- (ii) State one change in the design of the thermometer to increase its range.
-
- [1]
- (b) (i) Describe how the behaviour of a more sensitive thermometer is different from a less sensitive thermometer.
-
- [1]
- (ii) State one change in the design of the thermometer to make it more sensitive.
-
- [1]
- (c) Describe how a clinical thermometer differs from a laboratory thermometer. A diagram may be included in your answer.

.....

.....

.....

..... [3]

- (d) (i) In the space below, draw a labelled diagram of a thermocouple thermometer.

[2]

- (ii) State **two** reasons why a thermocouple thermometer is sometimes a better choice than a laboratory thermometer.

.....

 [2]

- (e) Fig. 9.2 shows a copper block of mass 1.8 kg with two holes in the top. An 80 W heater is placed in one hole and a thermometer in the other.

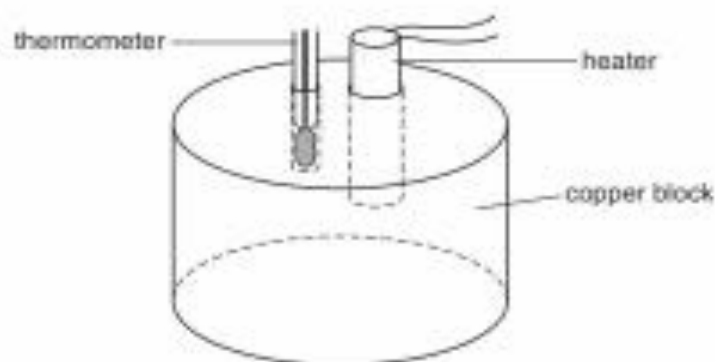


Fig. 9.2

The heater is switched on for 5.0 minutes. Assume that no energy is lost from the block.

- (i) Calculate the energy supplied to the block.

energy supplied = [2]

- (ii) The specific heat capacity of copper is $390 \text{ J/(kg}^\circ\text{C)}$. Calculate the rise in temperature of the block.

temperature rise = [2]

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MARK SCHEME

O/N18/21/Q5

5(a)(ii) large bulb / quantity of mercury

greater increase in volume

or

narrow bore

greater distance / difference (for a given increase in volume)

M1A1

5(b) traps the liquid above the constriction (so that the reading is maintained) **B1**

5(c) they / molecules gain kinetic energy / move faster **B1**

move apart **or** push each other apart **B1**

mercury expands (up the tube) **B1**

O/N17/21/Q4

4(a) use of boiling water **B1**

thermometer in boiling water or in steam above boiling water **B1**

mercury level at 100 °C mark **or** use of pure / distilled water **or** at a pressure of one atmosphere **B1**

4(b) (range is) decreased / smaller / reduced **B1**

ethanol (thread) reaches the end at a lower temperature

M/J17/21/Q11(c)

11(c)(i) two different metals connected together **B1**

voltmeter, ammeter or galvanometer in series **B1**

clear junction of two different metals used as the place to measure temperature **B1**

11(c)(ii) ANY 2 lines from

measures high(er) temperatures

more sensitive to changes in temperature

measures fast(er) changing temperatures / reading can be taken more rapidly

produce electronic output / output may be stored or processed / may be read remotely

O/N16/21/Q4

4 (a) greatest air; least copper **B1**

(b) (i) 1 difference between smallest and largest temperature **B1 or** from 0 to 100 °C

(i) 2 small/moderate distance between (thermometer) marks **B1**

or for a given temperature change there is a small expansion of liquid / distance(along scale) / change in thermometric property

or cannot measure small temperature difference / change

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- (ii) • use liquid that expands more B1
- smaller bore / thinner tube
- more mercury (in bulb) or use larger bulb

0/N15/22/Q5

- 5 (a) (i) (the property) varies with temperature B1
(ii) any two from:
volume (of gas / liquid) or density or length (of thread)
voltage or current or e.m.f.
resistance
pressure (of gas)
colour
(quantity of) radiation emitted
liquid crystal structure B2
(b) (i) temperature of melting ice / where water freezes or water / ice mixture B1
(ii) immerse thermometer in melting ice / at the ice point or boiling water / at steam point
or ice point and steam point marked / found (may be implied) B1
divide the difference into 100 units / sections B1 [6]

0/N13/21/Q5

- 5 (a) (thin-walled) bulb and capillary tube B1
mercury/liquid in bulb and constriction/U-bend B1
(b) mercury/liquid contracts B1
mercury/liquid/thread breaks (at the constriction)/constriction stops the mercury falling back
B1 [4]

0/N12/21/Q2

- 2 (a) (i) output/voltage/e.m.f. (directly) proportional to temperature (difference) B1
(ii)
800 750
7.70 6.20
—
—
or 1.5/50 or 0.03 or 0.6/1.5 or 20(°C) C1
770 °C A1
(b) glass melts/liquid boils/no remote reading (e.g. head in furnace) B1 [4]

0/N11/21/Q3

- 3 (a) (i) one junction in flame and three wires and fixed point/ice bath or two wires B1
two different metals and voltmeter connected B1
(ii) voltmeter reading/voltage at fixed points (e.g. V_0 and V_{100}) B1
compare V_{flame} with $V_{\text{fixed points}}$ (to obtain T) graph/equation/words B1

M/J10/21/Q9

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9 (a) (i) 120°C or -10°C to 110°C B1 [1]

(ii) longer thermometer or wider bore or less mercury or smaller bulb not

change liquid B1 [1]

(b) (i) measures small(er) change in temperature or small(er) range for same

distance or large(r) expansion for (same) temperature rise B1 [1]

(ii) larger bulb or more liquid or narrower bore/tube or use liquid that expands

more B1 [1]

(c) constriction/narrowing (accept 1st and 3rd marks on diagram)

mercury/thread breaks at constriction (on cooling) or thermometer is a “maximum” thermometer range different more sensitive/divisions further apart

triangular cross-section/acts as lens

thin(ner) bulb (quick response to temperature change) ANY 3 lines B3 [3]

(d) (i) two different metals joined M1

connected to meter/ammeter/galvanometer/voltmeter A1 [2]

(ii) low/high temperatures or greater range

responds quickly/measures rapidly changing temperatures

measures temperature at a point

electronic output

more robust measures temperatures at a distance (not more sensitive) ANY 2 lines B2 [2]

(e) (i) $(Q =) Pt$ or $80 \times 5 \times 60$ or 80×5 or 400 or 300 (s) seen C1

24000 J cao A1 [2]

(ii) $(Q =) mc\Delta T$ or $1.8 \times 390 \times T = 24000$ in any form ecf (i) C1

34°C (accept 34.188, 34.18, 34.19, 34.2) A1 [2]

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MARK SCHEME

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